The Journal of Open Source Software

TelescopeML – I. An End-to-End Python Package for Interpreting Telescope Datasets through Training Machine Learning Models, Generating Statistical Reports, and Visualizing Results

Ehsan (Sam) Gharib-Nezhad [1,2], **Natasha E. Batalha** [1], **Hamed Valizadegan** [3,4], **Miguel J. S. Martinho** [3,4], **Mahdi Habibi** [5], and **Gopal Nookula** [6]

1 Space Science and Astrobiology Division, NASA Ames Research Center, Moffett Field, CA, 94035 USA **2** Bay Area Environmental Research Institute, NASA Research Park, Moffett Field, CA 94035, USA **3** Universities Space Research Association (USRA), Mountain View, CA 94043, USA **4** Intelligent Systems Division, NASA Ames Research Center, Moffett Field, CA 94035, USA **5** Institute for Radiation Physics, Helmholtz-Zentrum Dresden-Rossendorf, Dresden 01328, Germany **6** Department of Computer Science, University of California, Riverside, Riverside, CA 92507 USA

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Paul La Plante

Reviewers:

- @oparisot
- @mwalmsley

Submitted: 29 November 2023
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

In partnership with

AMERICAN ASTRONOMICAL SOCIETY

This article and software are linked with research article DOI LINK OF ApJ PAPER - TBD, published in the The Astrophysical Journal.

DRAFT

Summary

We are on the verge of a revolutionary era in space exploration, thanks to advancements in telescopes such as the James Webb Space Telescope (*JWST*). High-resolution, high Signal-to-Noise spectra from exoplanet and brown dwarf atmospheres have been collected over the past few decades, requiring the development of accurate and reliable pipelines and tools for their analysis. Accurately and swiftly determining the spectroscopic parameters from the observational spectra of these objects is crucial for understanding their atmospheric composition and guiding future follow-up observations. `TelescopeML` is a Python package developed to perform three main tasks: (i) Process the synthetic astronomical datasets for training a CNN model and prepare the observational dataset for later use for prediction; (ii) Train a CNN model by implementing the optimal hyperparameters; and (iii) Deploy the trained CNN models on the actual observational data to derive the output spectroscopic parameters.

Statement of Need

We are on in a new era of space exploration, thanks to advancements in ground- and space-based telescopes, such as the James Webb Space Telescope (e.g., Gardner et al., 2023) and CRIRES. These remarkable instruments collect high-resolution, high-signal-to-noise spectra from extrasolar planets (e.g., Alderson et al., 2023), and brown dwarfs (e.g., Miles et al., 2023) atmospheres. Without an accurate interpretation of this data, the main objectives of space missions will not be fully accomplished. Different analytical and statistical methods, such as the chi-squared-test, Bayesian statistics as well as radiative-transfer atmospheric modeling packages have been developed (e.g., Batalha et al., 2019; MacDonald, 2023a) to interpret the spectra. They utilize either forward- and/or retrieval-radiative transfer modeling to analyze the spectra and extract physical information, such as atmospheric temperature, metallicity, carbon-to-oxygen ratio, and surface gravity (Iyer et al., 2023; Michael R. Line et al., 2014; M. S. Marley & Robinson, 2015). These atmospheric models rely on generating the physics and chemistry of these atmospheres for a wide range of thermal structures and compositions. In addition to Bayesian-based techniques, machine learning and deep learning methods have

been developed in recent years for various astronomical problems, including confirming the classification of light curves for exoplanet validation (e.g., Valizadegan et al., 2021), recognizing molecular features (Zingales & Waldmann, 2018) as well as interpreting brown dwarfs spectra using Random Forest technique (e.g., Lueber et al., 2023). Here, we present one of the first applications of deep learning and convolutional neural networks on the interpretation brown dwarf atmospheric datasets. The configuration of a CNN and the key concepts can be found in (Goodfellow et al., 2016; Kiranyaz et al., 2021).

With the continuous observation of these objects and the increasing amount of data, there is a critical need for a systematic pipeline to quickly explore the datasets and extract important physical from them. In the future we can expand our pipeline to exoplanet atmospheres, and use it to provide insights about the diversity of exoplanets and brown dwarfs' atmospheric compositions. Ultimately, `TelescopeML` will help facilitate the long-term analysis of this data in research. `TelescopeML` is an ML Python package with Sphinx-ed user-friendly documentation that provides both trained ML models and ML tools for interpreting observational data captured by telescopes.

Functionality and Key Features

`TelescopeML` is a Python package comprising a series of modules, each equipped with specialized machine learning and statistical capabilities for conducting Convolutional Neural Networks (CNN) or Machine Learning (ML) training on datasets captured from the atmospheres of extrasolar planets and brown dwarfs. The tasks executed by the `TelescopeML` modules are outlined below and visualized in following Figure:

- **DataMaster module**: Performs various tasks to process the datasets, including:
  - Load the training dataset (i.e., atmospheric fluxes) in CSV format
  - Split the dataset into training, validation, and test sets to pass it to the CNN model
  - Scale/normalize the dataset column-wise or row-wise
  - Visualize the training sets in each of the processing step for more insights
  - Perform feature engineering by extracting the Min and Max values from each flux to improve the ML training performance
- **DeepTrainer module**: Utilizes different methods/packages such as TensorFlow to:
  - Load the processed dataset from the **DataMaster** module
  - Build Convolutional Neural Networks (CNNs) model using the tuned hyperparameters
  - Fit/train the CNN models given the epochs, learning rate, and other parameters
  - Visualize the loss and training history, as well as the trained model's performance
- **Predictor module**: Implements the following tasks to predict atmospheric parameters:
  - Perform Scale/normalize processes on the observational fluxes
  - Deploys the trained CNNs model
  - Predict atmospheric parameters, i.e., effective temperature, gravity, carbon-to-oxygen ratio, and metallicity
  - Visualizes the processed observational dataset and the uncertainty in the predicted results
- **StatVisAnalyzer module**: Provides a set of functions to perform the following tasks:
  - Explores and processes the synthetic datasets
  - Performs the chi-square test to evaluate the similarity between two datasets
  - Calculates confidence intervals and standard errors
  - Functions t

DRAFT

The Journal of Open Source Software

Details on the synthetic dataset

The training dataset (or synthetic spectra) in this study is computed using the open-source atmospheric radiative transfer Python package, `PICASO` (e.g., Batalha et al., 2019), based on the `Sonora-Bobcat` model grid generated for cloudless brown dwarf atmospheres by (Mark S. Marley et al., 2021). This set encompasses 30,888 synthetic spectra, each including 104 wavelengths (i.e., 0.897, 0.906, ..., 2.512 μm) and their corresponding flux values. Each of these spectra has four output variables attached to it: effective temperature, gravity, carbon-to-oxygen ratio, and metallicity. These synthetic spectra are utilized to interpret observational datasets and derive these four atmospheric parameters. An example of the synthetic and observational dataset is shown in the following figure.

Details on the synthetic dataset

The training dataset (or synthetic spectra) in this study is computed using the open-source atmospheric radiative transfer Python package, `PICASO` (e.g., Batalha et al., 2019), based on the `Sonora-Bobcat` model grid generated for cloudless brown dwarf atmospheres by (Mark S. Marley et al., 2021). This set encompasses 30,888 synthetic spectra, each including 104 wavelengths (i.e., 0.897, 0.906, ..., 2.512 μm) and their corresponding flux values. Each of these spectra has four output variables attached to it: effective temperature, gravity, carbon-to-oxygen ratio, and metallicity. These synthetic spectra are utilized to interpret observational datasets and derive these four atmospheric parameters. An example of the synthetic and observational dataset is shown in the following figure.

Details on the CNN methodology for Multi-output Regression problem

Each row in the synthetic spectra has 104 input variables. The order of these data points and their magnitude are crucial to interpret the telescope data. For this purpose, we implemented a Convolutional Neural Network (CNN) method with 1-D convolutional layers. CNN is a powerful technique for this study because it extracts the dominant features from these spectra and then passes them to the fully connected hidden layers to learn the patterns. The output layer predicts the four atmospheric targets. An example of the CNN architecture is depicted in the following figure.

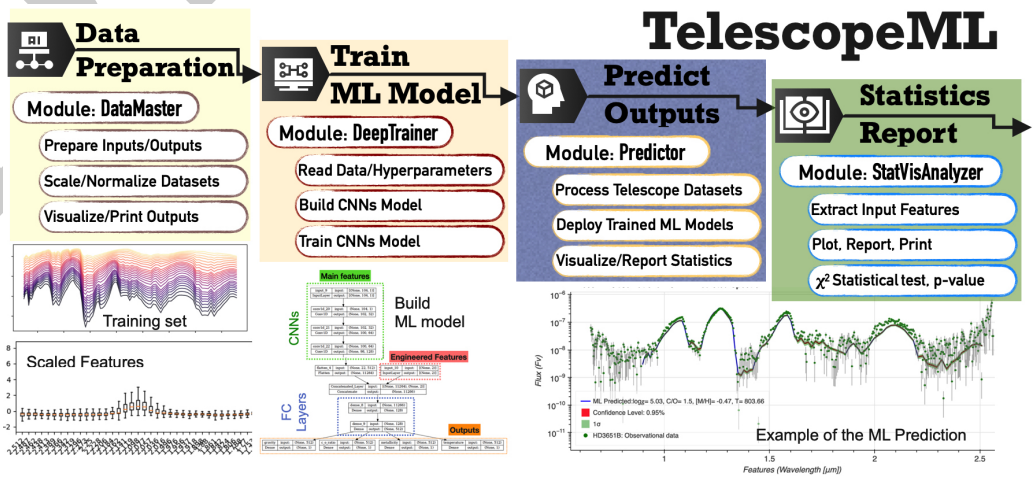

Figure 1: TelescopeML main modules to manipulate the training example, build the ML model, train and tune it, and ultimately extract the target features from the observational data.

DRAFT

Documentation

TelescopeML is available and being maintained as a GitHub repository at github.com/EhsanGharibNezhad/TelescopeML. Online documentation is hosted with *Sphinx* using *ReadtheDocs* tools and includes several instructions and tutorials as follows:

- **Main page**: ehsangharibnezhad.github.io/TelescopeML/
- **Installation**: ehsangharibnezhad.github.io/TelescopeML/installation.html
- **Tutorials and examples**: ehsangharibnezhad.github.io/TelescopeML/tutorials.html
- **The code**: ehsangharibnezhad.github.io/TelescopeML/code.html

Users and Future Developments

Astrophysicists with no prior machine learning knowledge can deploy the TelescopeML package and download the pre-trained ML or CNN models to interpret their observational data. In this scenario, pre-trained ML models, as well as the PyPI package, can be installed and deployed following the online instructions. Tutorials in the Sphinx documentation include examples for testing the code and also serve as a starting point. For this purpose, a basic knowledge of Python programming is required to install the code, run the tutorials, deploy the modules, and extract astronomical features from the datasets. The necessary machine learning background and a detailed guide for package installation, along with links to further Python details, are provided to help understand the steps and outputs.

Astrophysicists with machine learning expertise and data scientists can also benefit from this package by developing and fine-tuning the modules and pre-trained models to accommodate more complex datasets from various telescopes. This effort could also involve the utilization of new ML and deep learning algorithms, adding new capabilities such as feature engineering methods, and further optimization of hyperparameters using different and more efficient statistical techniques. The ultimate outcome from these two groups would be the creation of more advanced models with higher performance and robustness, as well as the extension of the package to apply to a wider range of telescope datasets.

Similar Tools

The following open-source tools are available to either perform forward modeling (χ^2-based test) or retrievals (based on Bayesian statistics and posterior distribution): Starfish (Czekala et al., 2015), petitRADTRANS (Mollière et al., 2019), petitRADTRANS (MacDonald, 2023b), PLATON (Zhang et al., 2019), CHIMERA (Michael R. Line et al., 2013), TauRex [Waldmann2015], NEMESIS (Irwin et al., 2008), and Pyrat Bay (Cubillos & Blecic, 2021).

In addition, the following package implements random forest to predict the atmospheric parameters: HELA (Márquez-Neila et al., 2018)

Utilized Underlying Packages

For processing datasets and training ML models in TelescopeML, the following software/packages are employed: Scikit-learn (Pedregosa et al., 2011), TensorFlow (Abadi et al., 2015), AstroPy (Astropy Collaboration et al., 2022), SpectRes (Carnall, 2017), Pandas (team, 2020), NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), Matplotlib (Hunter, 2007), Seaborn (Waskom, 2021), Bokeh (Bokeh Development Team, 2018). Additionally, for generating training astronomical datasets, Picaso (Batalha et al., 2019) is implemented.

DRAFT

Acknowledgements

EGN and GN would like to thank OSTEM internships and funding through the NASA with contract number 80NSSC22DA010. EGN acknowledges ChatGPT 3.5 for proofreading some of the functions. EGN is grateful to Olivier Parisot and Mike Walmsley for helpful referee reports, and to the JOSS editorial staff, Paul La Plante and Dan Foreman-Mackey, for their tireless efforts to encourage new people to join the open source community in astronomy.

References

Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., … Zheng, X. (2015). *TensorFlow: Large-scale machine learning on heterogeneous systems*. https://www.tensorflow.org/

Alderson, L., Wakeford, H. R., Alam, M. K., Batalha, N. E., Lothringer, J. D., Adams Redai, J., Barat, S., Brande, J., Damiano, M., Daylan, T., Espinoza, N., Flagg, L., Goyal, J. M., Grant, D., Hu, R., Inglis, J., Lee, E. K. H., Mikal-Evans, T., Ramos-Rosado, L., … Zhang, X. (2023). Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H. *Nature*, *614*(7949), 664–669. https://doi.org/10.1038/s41586-022-05591-3

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., Earl, N., Starkman, N., Bradley, L., Shupe, D. L., Patil, A. A., Corrales, L., Brasseur, C. E., N"othe, M., Donath, A., Tollerud, E., Morris, B. M., Ginsburg, A., Vaher, E., Weaver, B. A., Tocknell, J., Jamieson, W., … Astropy Project Contributors. (2022). The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package. *935*(2), 167. https://doi.org/10.3847/1538-4357/ac7c74

Batalha, N. E., Marley, M. S., Lewis, N. K., & Fortney, J. J. (2019). Exoplanet reflected-light spectroscopy with PICASO. *The Astrophysical Journal*, *878*(1), 70. https://doi.org/10.3847/1538-4357/ab1b51

Bokeh Development Team. (2018). *Bokeh: Python library for interactive visualization*. https://bokeh.pydata.org/en/latest/

Carnall, A. C. (2017). SpectRes: A Fast Spectral Resampling Tool in Python. *arXiv e-Prints*, arXiv:1705.05165. https://doi.org/10.48550/arXiv.1705.05165

Cubillos, P. E., & Blecic, J. (2021). The PYRAT BAY framework for exoplanet atmospheric modelling: a population study of Hubble/WFC3 transmission spectra. *Monthly Notices of the Royal Astronomical Society*, *505*(2), 2675–2702. https://doi.org/10.1093/mnras/stab1405

Czekala, I., Andrews, S. M., Mandel, K. S., Hogg, D. W., & Green, G. M. (2015). Constructing a Flexible Likelihood Function for Spectroscopic Inference. *812*(2), 128. https://doi.org/10.1088/0004-637X/812/2/128

Gardner, J. P., Mather, J. C., Abbott, R., Abell, J. S., Abernathy, M., Abney, F. E., Abraham, J. G., Abraham, R., Abul-Huda, Y. M., Acton, S., & al., et. (2023). The James Webb Space Telescope Mission. *135*(1048), 068001. https://doi.org/10.1088/1538-3873/acd1b5

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

DRAFT

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Irwin, P. G. J., Teanby, N. A., de Kok, R., Fletcher, L. N., Howett, C. J. A., Tsang, C. C. C., Wilson, C. F., Calcutt, S. B., Nixon, C. A., & Parrish, P. D. (2008). The NEMESIS planetary atmosphere radiative transfer and retrieval tool. *Journal of Quantitative Spectroscopy and Radiative Transfer*, *109*, 1136–1150. https://doi.org/10.1016/j.jqsrt.2007.11.006

Iyer, A. R., Line, M. R., Muirhead, P. S., Fortney, J. J., & Gharib-Nezhad, E. (2023). The SPHINX M-dwarf Spectral Grid. I. Benchmarking New Model Atmospheres to Derive Fundamental M-dwarf Properties. *944*(1), 41. https://doi.org/10.3847/1538-4357/acabc2

Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., & Inman, D. J. (2021). 1D convolutional neural networks and applications: A survey. *Mechanical Systems and Signal Processing*, *151*, 107398. https://doi.org/10.1016/j.ymssp.2020.107398

Line, Michael R., Knutson, H., Wolf, A. S., & Yung, Y. L. (2014). A systematic retrieval analysis of secondary eclipse spectra. II. A uniform analysis of nine planets and their c to o ratios. *The Astrophysical Journal*, *783*(2), 70.

Line, Michael R., Wolf, A. S., Zhang, X., Knutson, H., Kammer, J. A., Ellison, E., Deroo, P., Crisp, D., & Yung, Y. L. (2013). A Systematic Retrieval Analysis of Secondary Eclipse Spectra. I. A Comparison of Atmospheric Retrieval Techniques. *Astrophysical Journal*, *775*(2), 137. https://doi.org/10.1088/0004-637X/775/2/137

Lueber, A., Kitzmann, D., Fisher, C. E., Bowler, B. P., Burgasser, A. J., Marley, M., & Heng, K. (2023). Intercomparison of Brown Dwarf Model Grids and Atmospheric Retrieval Using Machine Learning. *954*(1), 22. https://doi.org/10.3847/1538-4357/ace530

MacDonald, R. J. (2023b). POSEIDON: A multidimensional atmospheric retrieval code for exoplanet spectra. *Journal of Open Source Software*, *8*(81), 4873. https://doi.org/10.21105/joss.04873

MacDonald, R. J. (2023a). POSEIDON: A multidimensional atmospheric retrieval code for exoplanet spectra. *Journal of Open Source Software*, *8*(81), 4873. https://doi.org/10.21105/joss.04873

Marley, M. S., & Robinson, T. D. (2015). On the Cool Side: Modeling the Atmospheres of Brown Dwarfs and Giant Planets. *53*, 279–323. https://doi.org/10.1146/annurev-astro-082214-122522

Marley, Mark S., Saumon, D., Visscher, C., Lupu, R., Freedman, R., Morley, C., Fortney, J. J., Seay, C., Smith, A. J., Teal, D., & others. (2021). The sonora brown dwarf atmosphere and evolution models. I. Model description and application to cloudless atmospheres in rainout chemical equilibrium. *The Astrophysical Journal*, *920*(2), 85.

Márquez-Neila, P., Fisher, C., Sznitman, R., & Heng, K. (2018). Supervised machine learning for analysing spectra of exoplanetary atmospheres. *Nature Astronomy*, *2*, 719–724. https://doi.org/10.1038/s41550-018-0504-2

Miles, B. E., Biller, B. A., Patapis, P., Worthen, K., Rickman, E., Hoch, K. K. W., Skemer, A., Perrin, M. D., Whiteford, N., Chen, C. H., Sargent, B., Mukherjee, S., Morley, C. V., Moran, S. E., Bonnefoy, M., Petrus, S., Carter, A. L., Choquet, E., Hinkley, S., … Zhang, Z. (2023). The JWST Early-release Science Program for Direct Observations of Exoplanetary Systems II: A 1 to 20 μm Spectrum of the Planetary-mass Companion VHS 1256-1257 b. *ApJL*, *946*(1), L6. https://doi.org/10.3847/2041-8213/acb04a

Mollière, P., Wardenier, J. P., van Boekel, R., Henning, Th., Molaverdikhani, K., & Snellen, I. A. G. (2019). petitRADTRANS. A Python radiative transfer package for exoplanet characterization and retrieval. *Astronomy & Astrophysics*, *627*, A67. https://doi.org/10.1051/0004-6361/201935470

DRAFT

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

team, T. pandas development. (2020). *Pandas-dev/pandas: pandas* (latest). Zenodo. https://doi.org/10.5281/zenodo.3509134

Valizadegan, H., Martinho, M., Wilkens, L. S., Jenkins, J. M., Smith, J., Caldwell, D. A., Twicken, J. D., Gerum, P. C., Walia, N., Hausknecht, K., Lubin, N. Y., Bryson, S. T., & Oza, N. C. (2021). ExoMiner: A Highly Accurate and Explainable Deep Learning Classifier that Validates 301 New Exoplanets. *arXiv e-Prints*, arXiv:2111.10009. https://arxiv.org/abs/2111.10009

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Waskom, M. L. (2021). Seaborn: Statistical data visualization. *Journal of Open Source Software*, *6*(60), 3021. https://doi.org/10.21105/joss.03021

Zhang, M., Chachan, Y., Kempton, E. M.-R., & Knutson, H. A. (2019). Forward Modeling and Retrievals with PLATON, a Fast Open-source Tool. *Publications of the Astronomical Society of the Pacific*, *131*(997), 034501. https://doi.org/10.1088/1538-3873/aaf5ad

Zingales, T., & Waldmann, I. P. (2018). ExoGAN: Retrieving Exoplanetary Atmospheres Using Deep Convolutional Generative Adversarial Networks. *156*(6), 268. https://doi.org/10.3847/1538-3881/aae77c

DRAFT